

Sales Data Cleaning and Visualization Project

Objective: Clean a raw dataset, handle missing values, detect outliers, and create visualizations to extract insights.

Tools Used: Python, Pandas, Matplotlib, Seaborn

Import Libraries

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")

print("Libraries Loaded")
```

Libraries Loaded

Load Dataset

```
In [3]: df = pd.read_csv(r"C:\Users\moham\OneDrive\Desktop\data_science_project\data/tips.c
df.head()
```

```
Out[3]:
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Data Understanding

```
In [4]: df.shape #shape of dataset
```

```
Out[4]: (244, 7)
```

In [5]: `df.columns` *#columns present in dataset*

Out[5]: Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='object')

In [6]: `df.info()` *#data types*

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   total_bill  244 non-null    float64
1   tip         244 non-null    float64
2   sex         244 non-null    object
3   smoker      244 non-null    object
4   day         244 non-null    object
5   time        244 non-null    object
6   size        244 non-null    int64
dtypes: float64(2), int64(1), object(4)
memory usage: 13.5+ KB
```

In [8]: `df.describe()` *#summary*

Out[8]:

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

Check Missing Values

In [9]: `df.isnull().sum()`

Out[9]:

total_bill	0
tip	0
sex	0
smoker	0
day	0
time	0
size	0
dtype:	int64

```
In [13]: df.ffmpeg(inplace=True) #fill the missing value
```

Check for duplicates

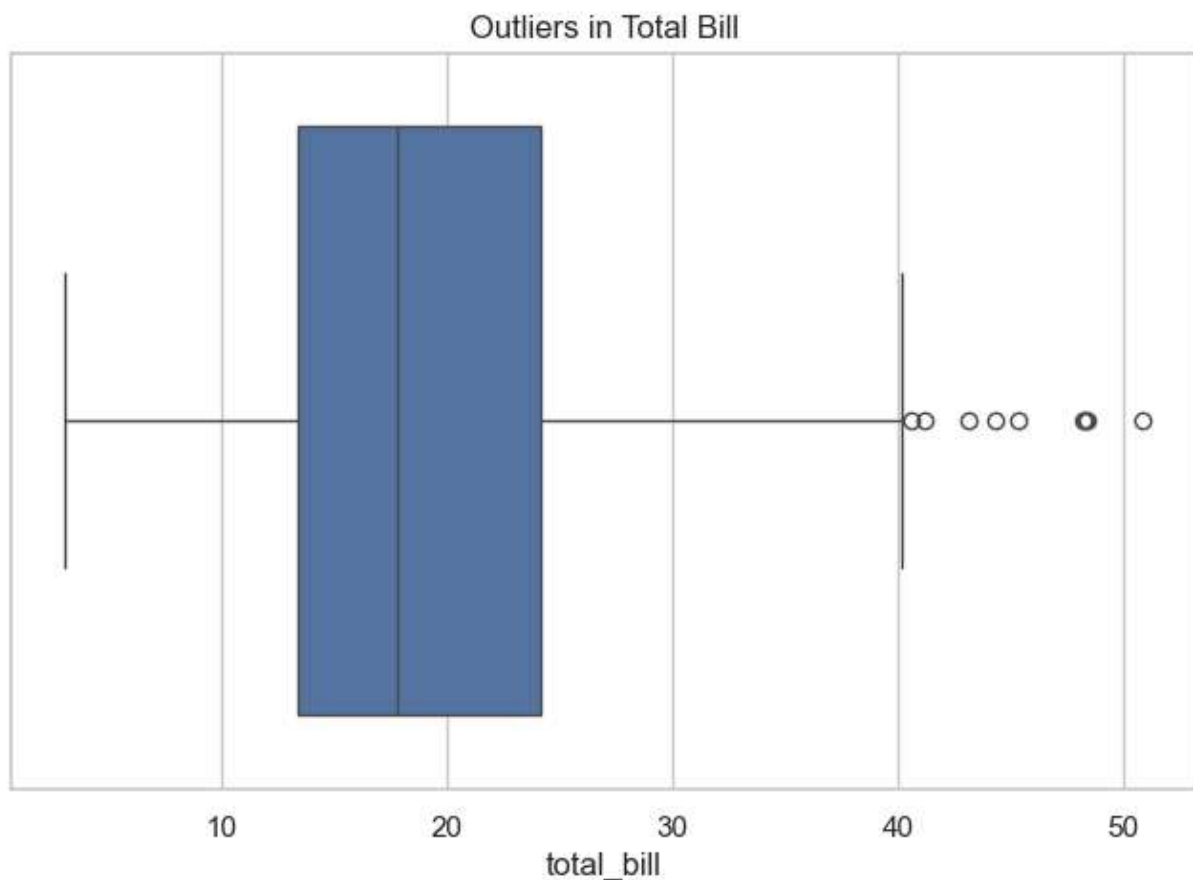
```
In [14]: df.duplicated().sum()
```

```
Out[14]: np.int64(1)
```

```
In [15]: df.drop_duplicates(inplace=True) #if duplicated exists delete them
```

Detect Outliers using boxplots

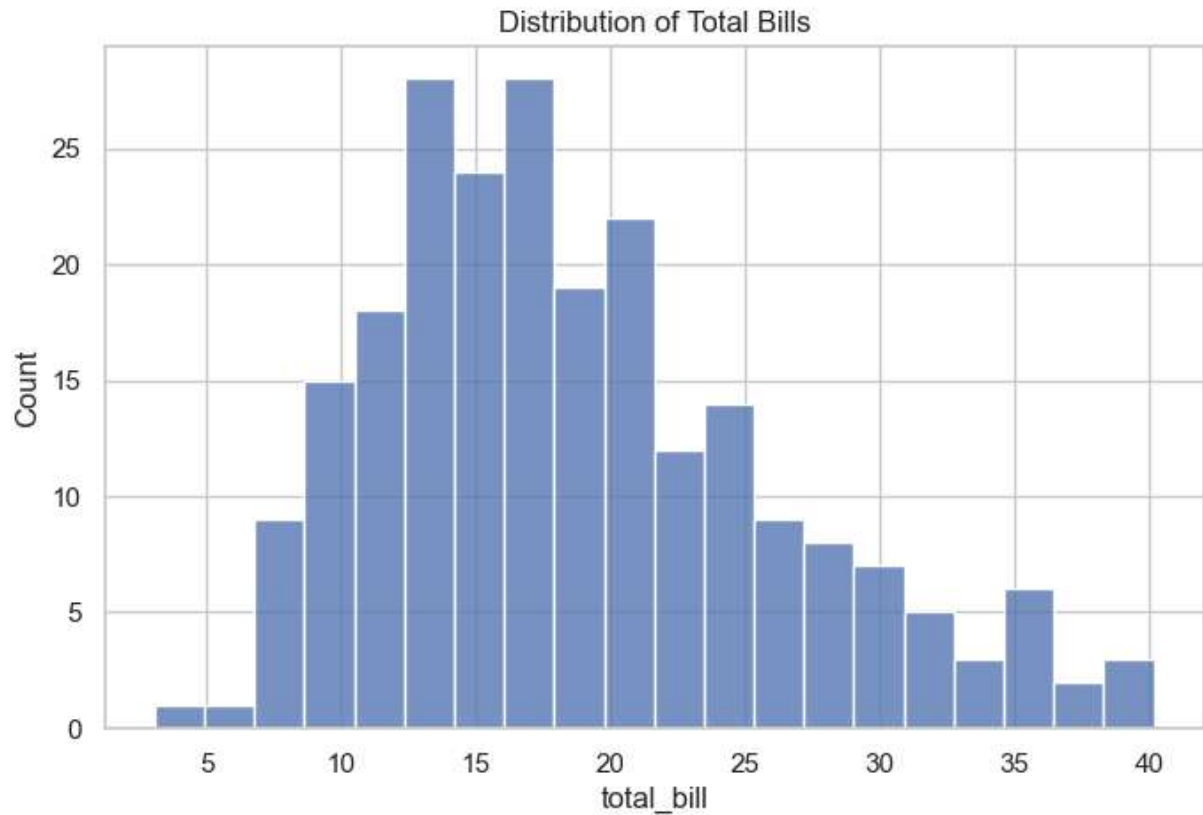
```
In [16]: plt.figure(figsize=(8,5))  
sns.boxplot(x=df["total_bill"])  
plt.title("Outliers in Total Bill")  
plt.show()
```



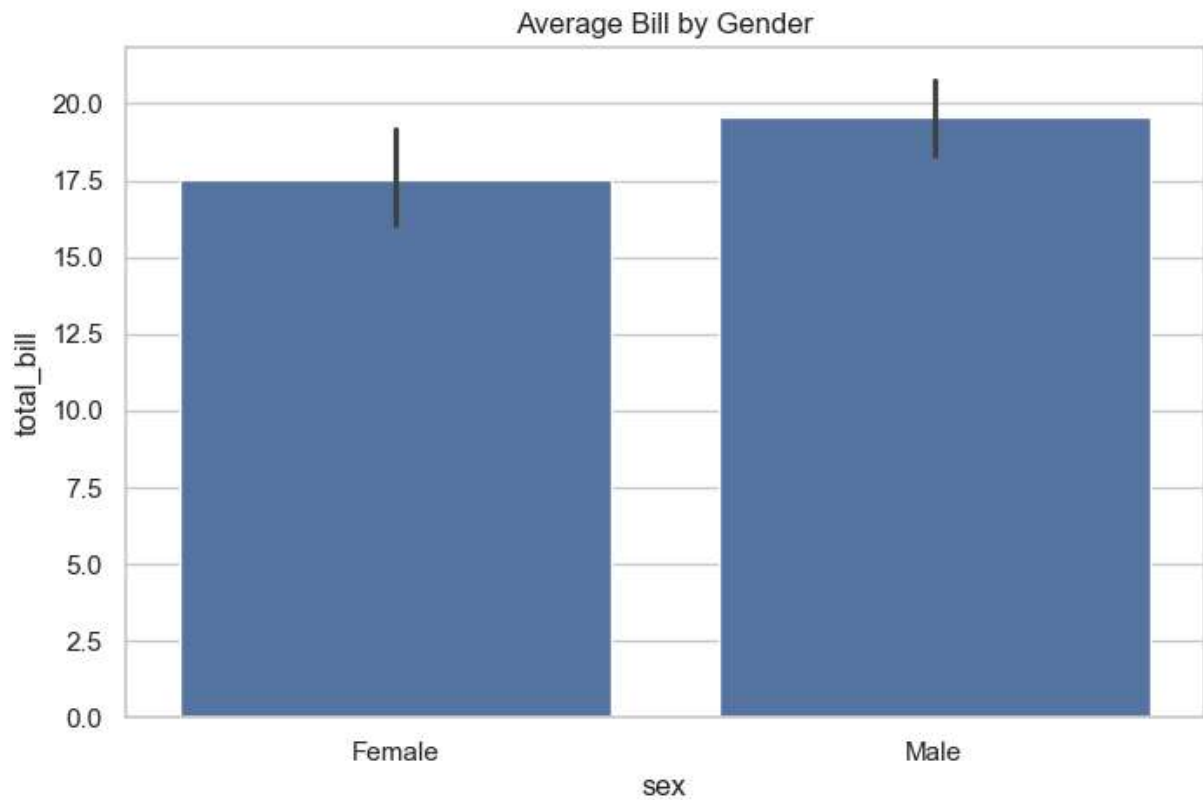
```
In [17]: Q1 = df["total_bill"].quantile(0.25)  
Q3 = df["total_bill"].quantile(0.75)  
IQR = Q3 - Q1  
df = df[(df["total_bill"] >= Q1 - 1.5*IQR) &  
        (df["total_bill"] <= Q3 + 1.5*IQR)] #remove outliers
```

Data Visualization

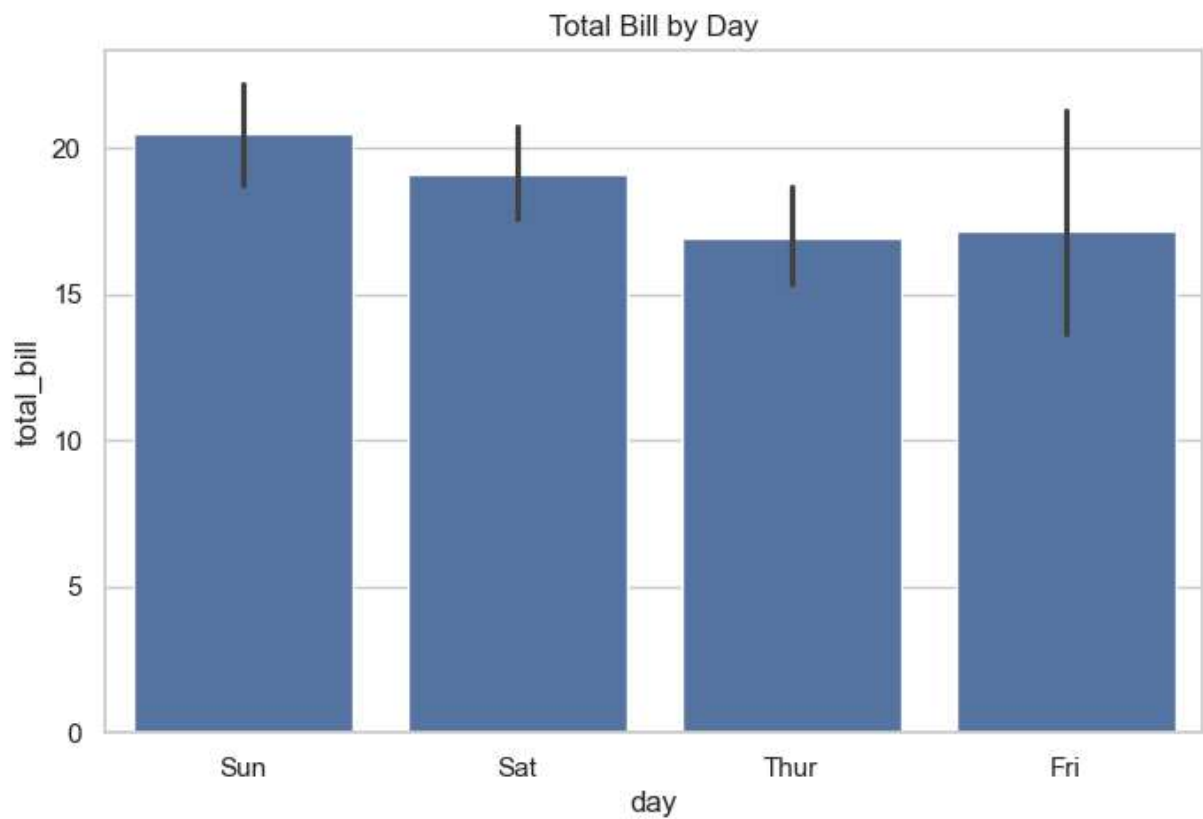
```
In [18]: #bill distribution using histograms
plt.figure(figsize=(8,5))
sns.histplot(df["total_bill"], bins=20)
plt.title("Distribution of Total Bills")
plt.show()
```



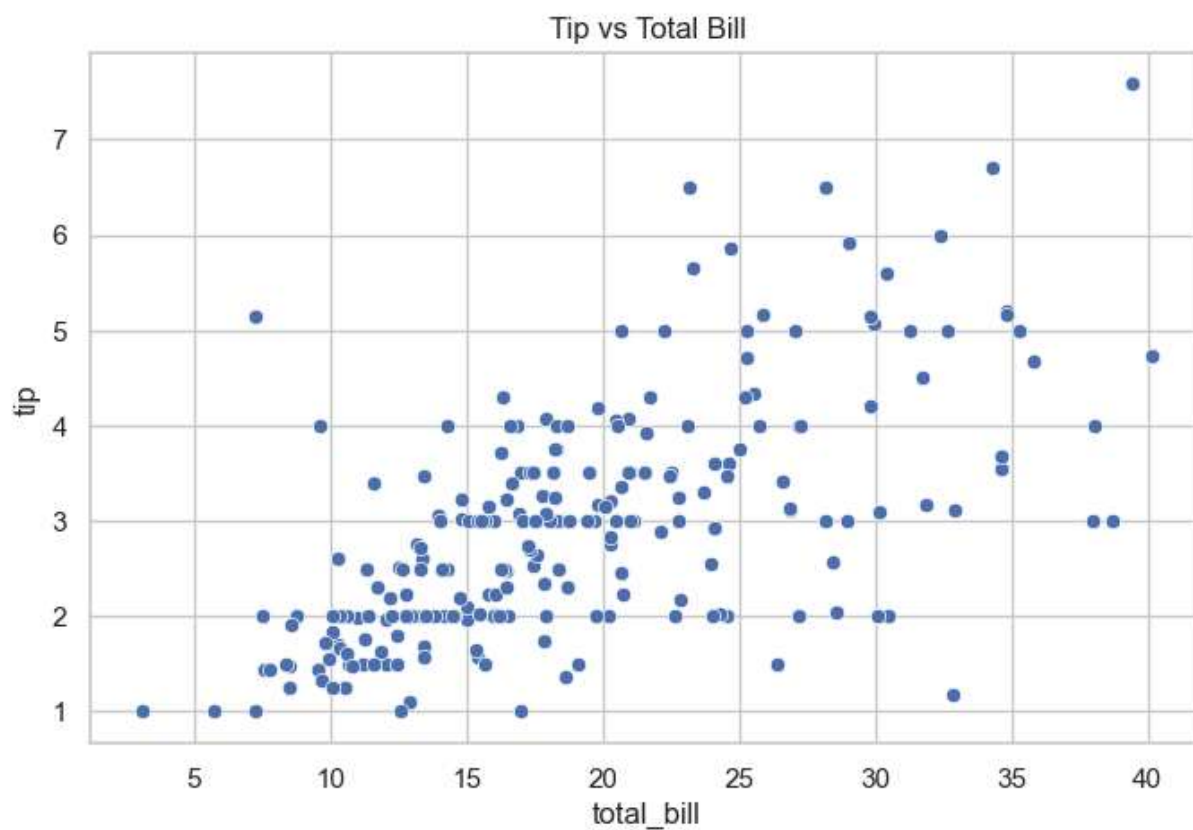
```
In [19]: #gender vs spending using bargraph
plt.figure(figsize=(8,5))
sns.barplot(x="sex", y="total_bill", data=df)
plt.title("Average Bill by Gender")
plt.show()
```



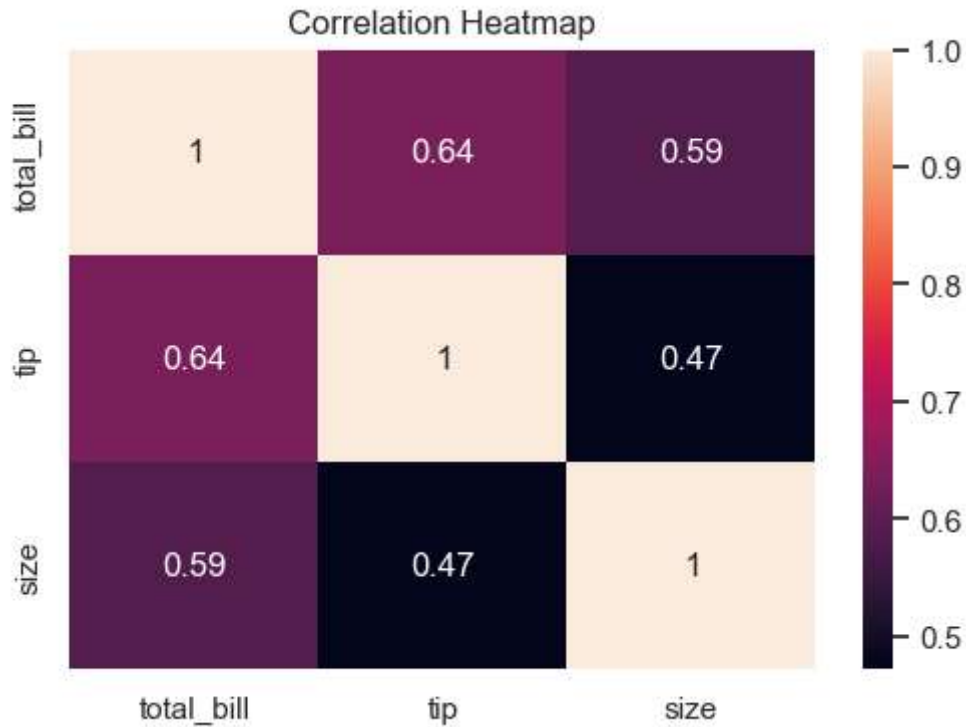
```
In [20]: #revenue vs day using bargraph
plt.figure(figsize=(8,5))
sns.barplot(x="day", y="total_bill", data=df)
plt.title("Total Bill by Day")
plt.show()
```



```
In [21]: #tip vs bill using scatter plot
plt.figure(figsize=(8,5))
sns.scatterplot(x="total_bill", y="tip", data=df)
plt.title("Tip vs Total Bill")
plt.show()
```



```
In [22]: #heatmap for correlation
plt.figure(figsize=(6,4))
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```



Key Insights

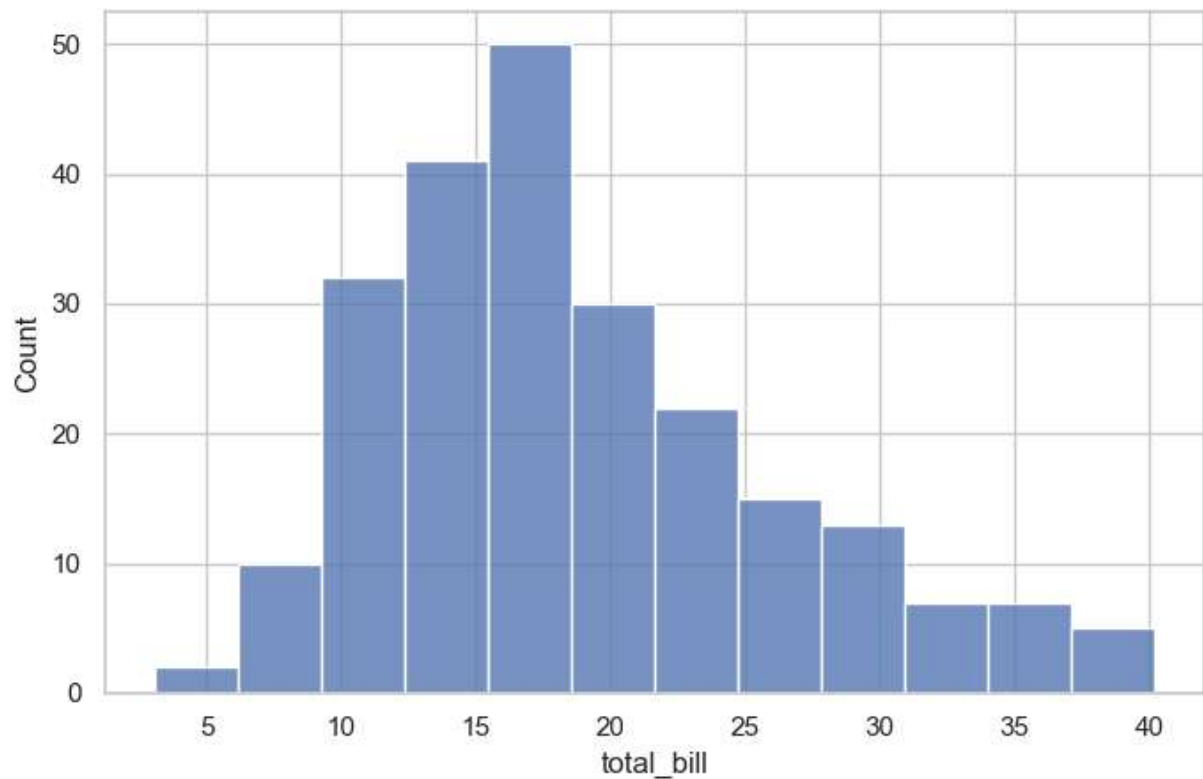
1. Total bill and tip are positively correlated.
2. Weekend days show higher spending.
3. Male customers have slightly higher average bills.
4. Higher bills usually result in larger tips.

Save Clean Dataset

```
In [23]: df.to_csv(r"C:\Users\moham\OneDrive\Desktop\data_science_project\data/cleaned_tips.
```

Save Visualizations

```
In [24]: plt.figure(figsize=(8,5))  
sns.histplot(df["total_bill"])  
plt.savefig(r"C:\Users\moham\OneDrive\Desktop\data_science_project\outputs/bill_dis  
plt.show()
```



Conclusion

This project performed data cleaning and exploratory data analysis on the tips dataset.

Steps performed: ~Handled missing values ~Removed duplicate records ~Detected and removed outliers ~Created visualizations to analyze patterns

The analysis shows that total bill strongly influences tip amount and weekend days generate higher spending.

In []: